The neutron dripline in calcium isotopes from a chiral interaction

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Paper discussion: motivation, equations, all figures and tables

Central question: how many neutrons can calcium bind?

Theoretical prediction : a two-neutron dripline near ^{71}Ca.

Separation energies and the dripline

Calcium has Z = 20 protons. The mass number is A = Z + N.

$$B(A,Z) = -E_{\mathrm{gs}}(A,Z)$$

$$S_{1n}(A,Z) = E_{\mathrm{gs}}(A-1,Z) - E_{\mathrm{gs}}(A,Z)$$

$$S_{2n}(A,Z) = E_{\mathrm{gs}}(A-2,Z) - E_{\mathrm{gs}}(A,Z)$$

$$S_{2n}(A,Z) = S_{1n}(A,Z) + S_{1n}(A-1,Z)$$

Egs uses the free-nucleon threshold as zero. B is binding energy; S_1n and S_2n are neutron-removal energies.

S > 0: bound against that channel. S < 0: the emission channel is open.

Equation (1): the intrinsic Hamiltonian

Remove the motion of the nucleus as a whole to describe its internal dynamics.

$$\hat{H} = \hat{T} - \hat{T}_{\mathrm{CoM}} + \hat{V}_{2N} + \hat{V}_{3N} \tag{1}$$

T: total kinetic energy. TCoM: center-of-mass kinetic energy. V_2N and V_3N: two- and three-nucleon interactions.

$$\hat{T} = \sum\nolimits_i \frac{\mathbf{p}_i^{\,2}}{2m}, \qquad \hat{T}_{\mathrm{CoM}} = \frac{\left(\sum_i \mathbf{p}_i\right)^2}{2Am}$$

$$\hat{H}|\Psi\rangle = E|\Psi\rangle$$

Ψ is the many-nucleon state and E its energy. Three-body forces also affect saturation and shell structure.

The interaction: chiral orders and regulation

$$V_{2N} : \mathrm{N^3LO}, \qquad V_{3N} : \mathrm{NNLO} = \mathrm{N^2LO}$$

The two-body and three-body sectors stop at different orders and share a smooth regulator:

$$f(p) = \exp\left[-\left(\frac{p}{\Lambda}\right)^{2n}\right], \qquad \Lambda = 394\,\mathrm{MeV}, \quad n = 4$$

p is momentum and Λ the cutoff. With 2n = 8, low momenta are retained and high momenta are strongly suppressed.

Fixed πN constants; 26 two-body contacts plus cD and cE give 28 optimized parameters.

Many-body methods and model spaces

J-NCSM for A ≤ 4; VS-IMSRG for O and Ca; CC for calcium and closed-shell tests.

$$e = 2n_r + \ell \le e_{\max}, \quad E_{\mathrm{sp}}^{\max} = (e_{\max} + 3/2)\hbar\omega$$

$$e_1 + e_2 + e_3 \le E_{3\,\max}, \quad e_{\max} = 14, \quad E_{3\,\max} = 28, \quad \hbar\omega = 12\ \mathrm{MeV}$$

emax sets the single-particle cutoff; E_3max truncates three-body matrix elements. nr and ℓ are radial and orbital quantum numbers.

Ca valence space: a ^{48}Ca core + {$1p_{3/2}$, $1p_{1/2}$, $0f_{5/2}$, $0g_{9/2}$, $1d_{5/2}$, $2s_{1/2}$}.

$$E_{\mathrm{used}} \approx E_{\mathrm{CCSD}} + 0.10(E_{\mathrm{CCSD}} - E_{\mathrm{HF}}) + \Delta E_{\mathrm{proj}}$$

Parameter fitting, selection, and prediction

2000 starts → 20 candidates → selection by 4 energies → N³LOTexas

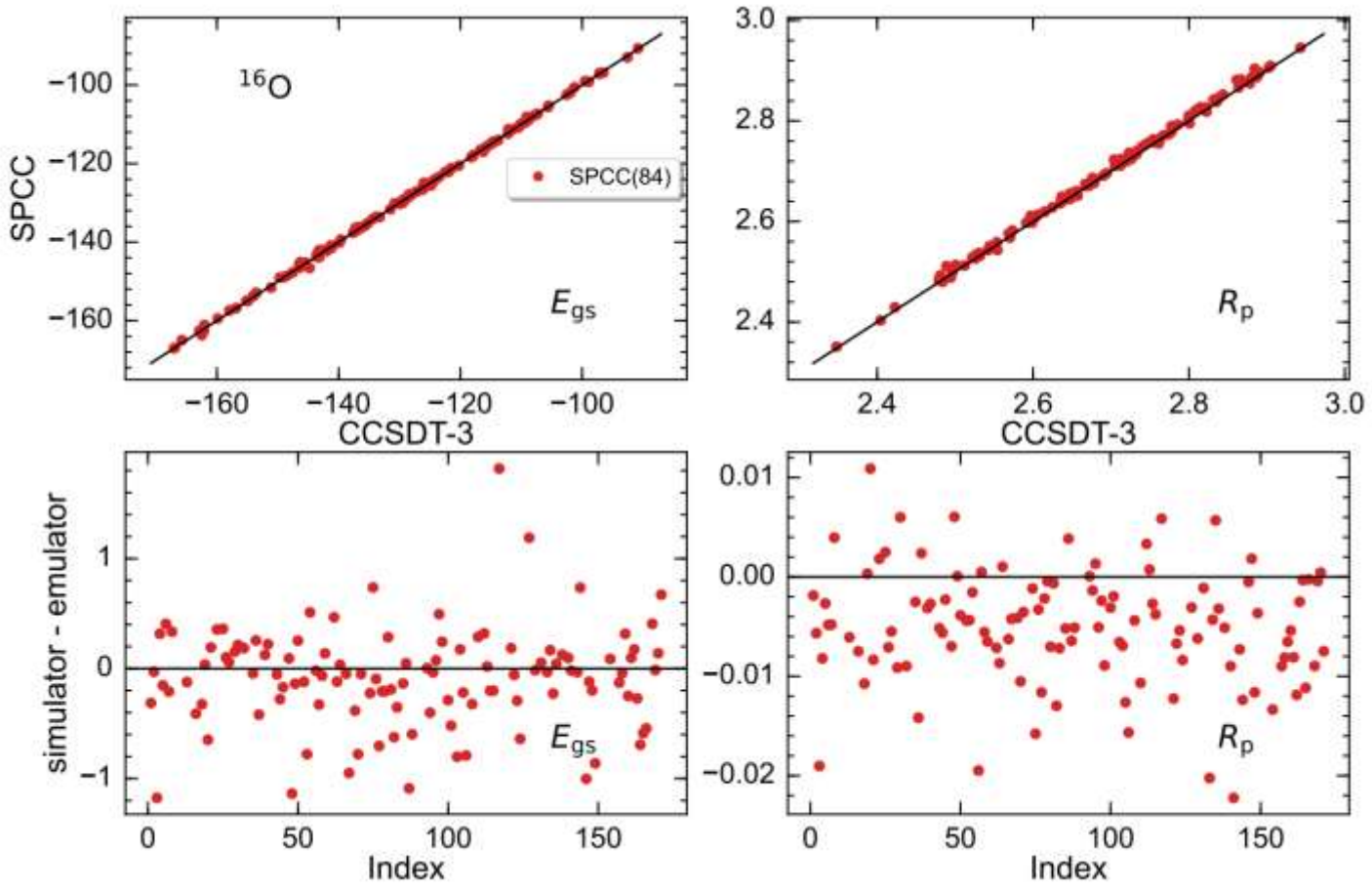
Fitting targets: np phase shifts, $^1S_0$ effective-range parameters, and selected ^{2}H, ^{4}He, and ^{16}O properties.

Final selection: ground-state energies of ^{22}O, ^{24}O, ^{40}Ca, and ^{48}Ca.

Remote driplines, other spectra, heavy nuclei, and nuclear matter provide further tests.

Figure S2: emulator versus full calculation

$$H_{\mathrm{sub}}(\boldsymbol{\theta})\mathbf{c} = E\,N_{\mathrm{sub}}\mathbf{c}$$



Fast emulator

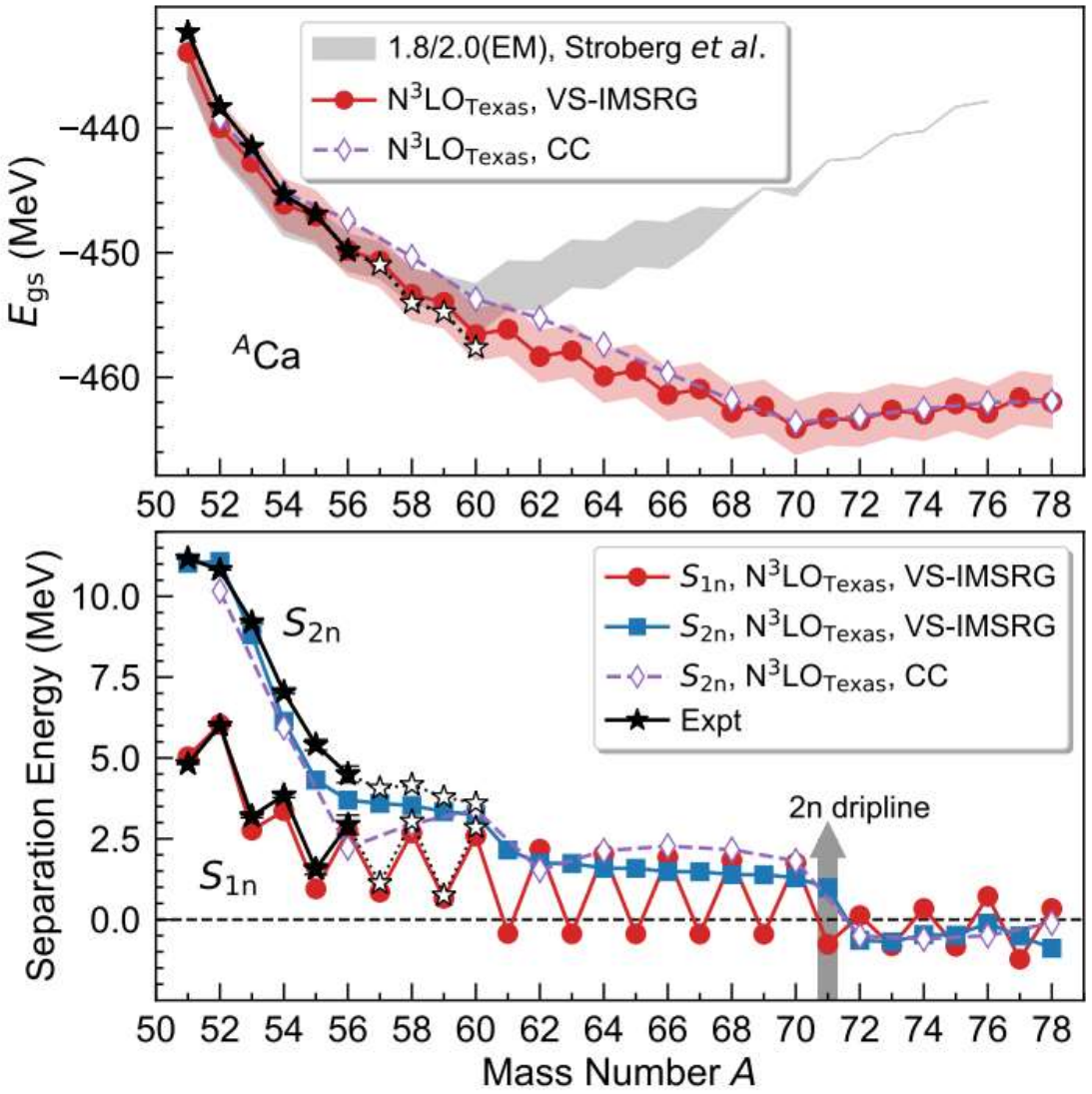
84 training points form an 84 × 84 problem, tested against 200 full CCSDT-3 results.

Equality and residuals

Top: close to y = x. Bottom: full calculation minus emulator. Right: point-proton radius Rp.

<1% is emulator discrepancy. The equation is a schematic projection; Nsub is the overlap matrix.

Figure 1: the two-neutron dripline near ^{71}Ca



Energy: continued binding

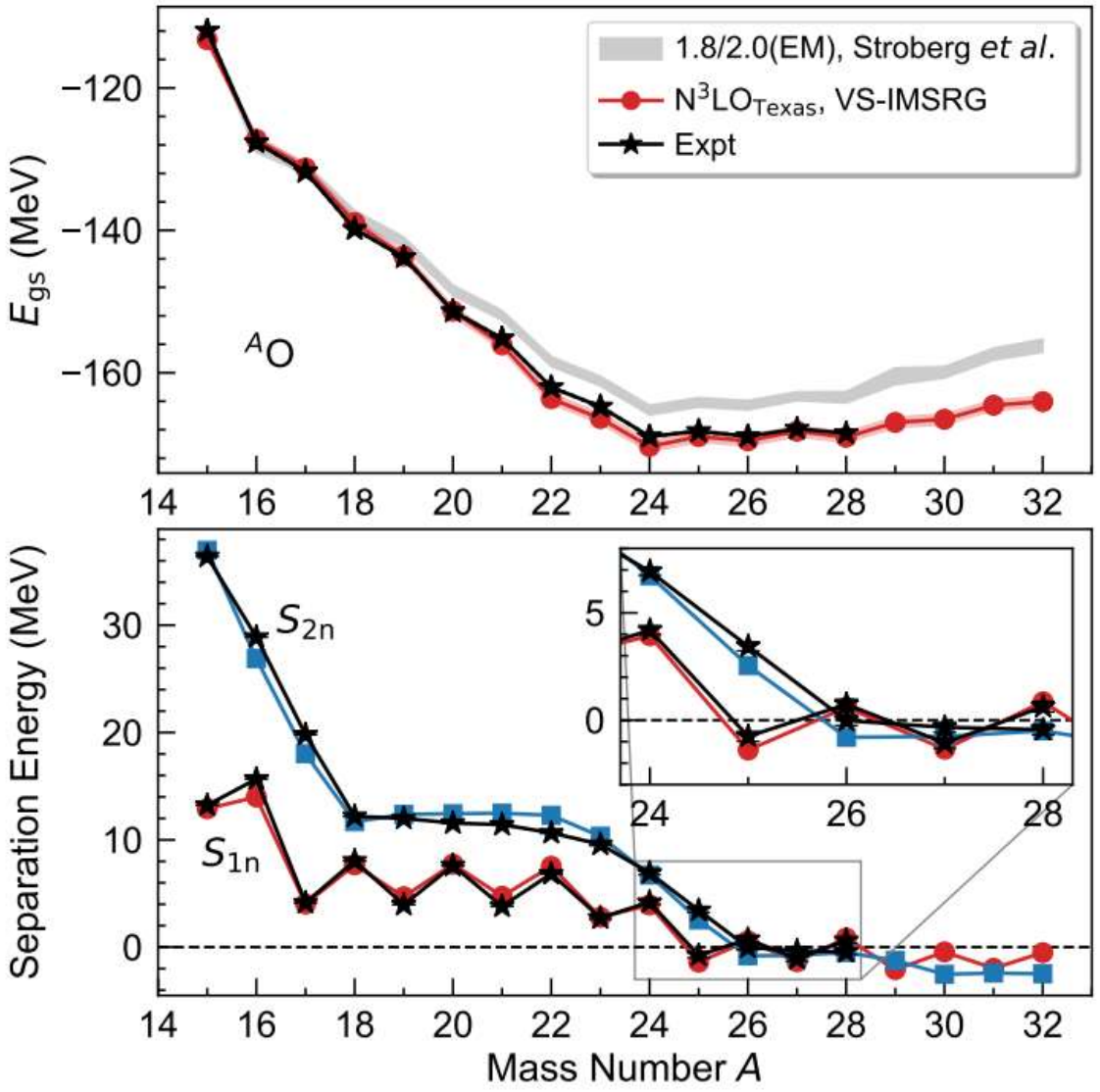
The new interaction continues to lower the energy beyond A = 60, flattening near A = 70.

Separation: an extended threshold

S_1n shows odd–even staggering; S_2n crosses zero near A = 70–72. CC and VS-IMSRG support the trend.

^{71}Ca labels the two-neutron threshold and has $S_1n < 0$ here. The even-A boundary is near ^{70}Ca.

Figure 2: oxygen still terminates near ^{24}O



Oxygen still ends near ^{24}O

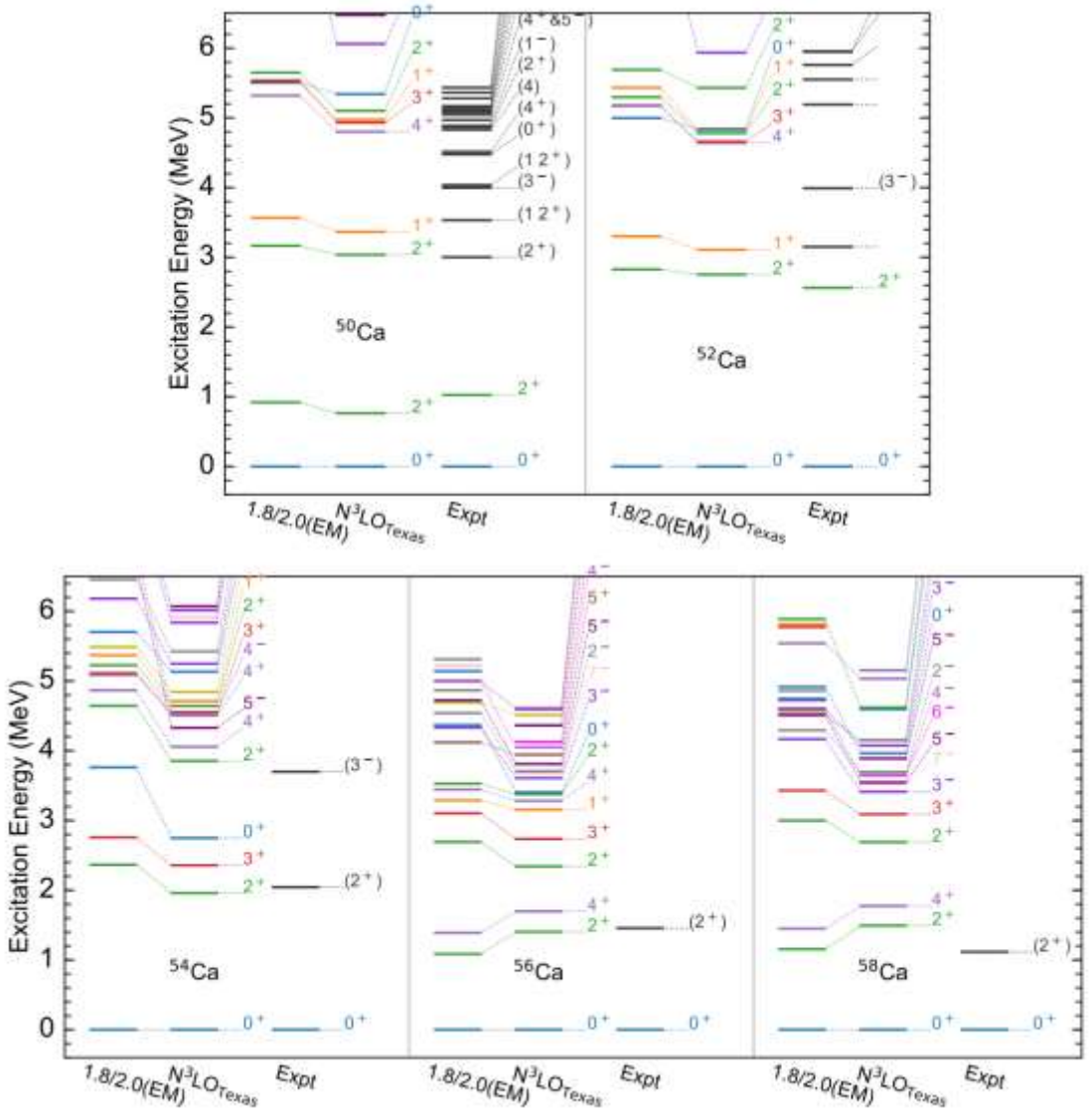
Energies and separation energies broadly follow the data. Heavier isotopes enter the unbound region near threshold.

Near-threshold sensitivity

Experimental context in the paper: ^{25}O lies about 830 keV above ^{24}O+n, and ^{26}O about 18 keV above ^{24}O+2n.

^{22}O and ^{24}O energies entered selection. Broad agreement does not establish precision for every resonance.

Figure S1: excitation spectra and shell structure



$$E_x(J^\pi) = E(J^\pi) - E_{\mathrm{gs}}$$

$$\pi = (-1)^\ell$$

Ex is energy above the ground state. Jπ gives angular momentum and parity. The lower equation applies to one orbital.

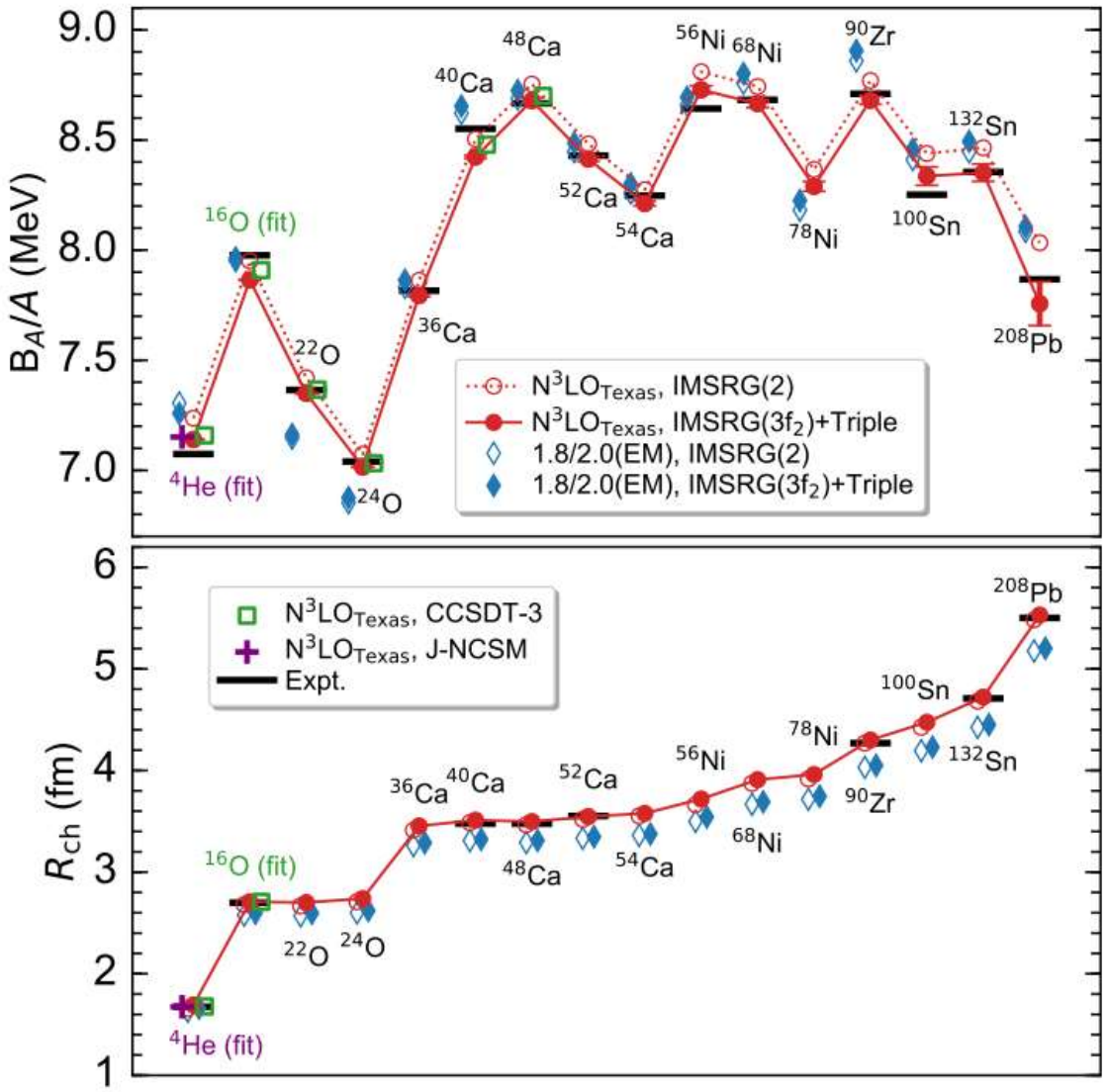
Added Ex: DOE-HDBK-1019/1-93, NP-01, p. 20.
Orbital parity: MIT 8.321 (2017), Lec. 22, p. 93.

Lower negative-parity states

Consistent with a smaller N = 40 subshell gap and easier cross-shell correlations.

Positive-parity spectra remain broadly comparable. The first 2^+ state in ^{58}Ca does not improve; a single Ex is not the shell gap.

Figure 3: binding energies and charge radii



$$B/A = -E_{\mathrm{gs}}/A$$

$$R_{\mathrm{ch}} = \sqrt{\langle r^2 \rangle_{\mathrm{ch}}}$$

Added B/A identity and rms-radius definition:
Hjorth-Jensen, TALENT (2015), “Binding energies” / “Radii”.

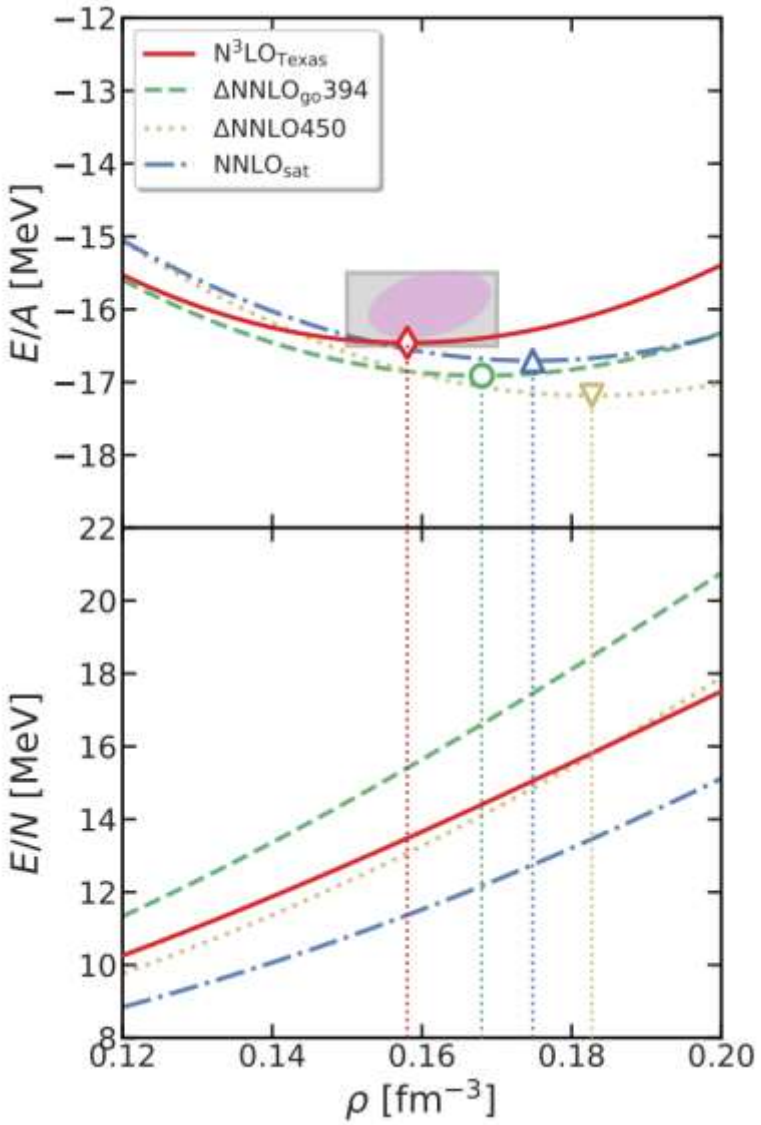
Energy and size constrain the force

Top: good binding-energy systematics retained. Bottom: improved charge radii relative to the undersized older predictions.

Rch is the rms charge radius. ^{4}He and ^{16}O entered fitting; “Triple” denotes perturbative triples.

The ≈1.5% accuracy concerns the bulk properties discussed.

Figure 4: nuclear-matter saturation



$$\rho = A/V, \qquad e(\rho) = E/A$$

$$e'(\rho_0) = 0, \qquad e''(\rho_0) > 0$$

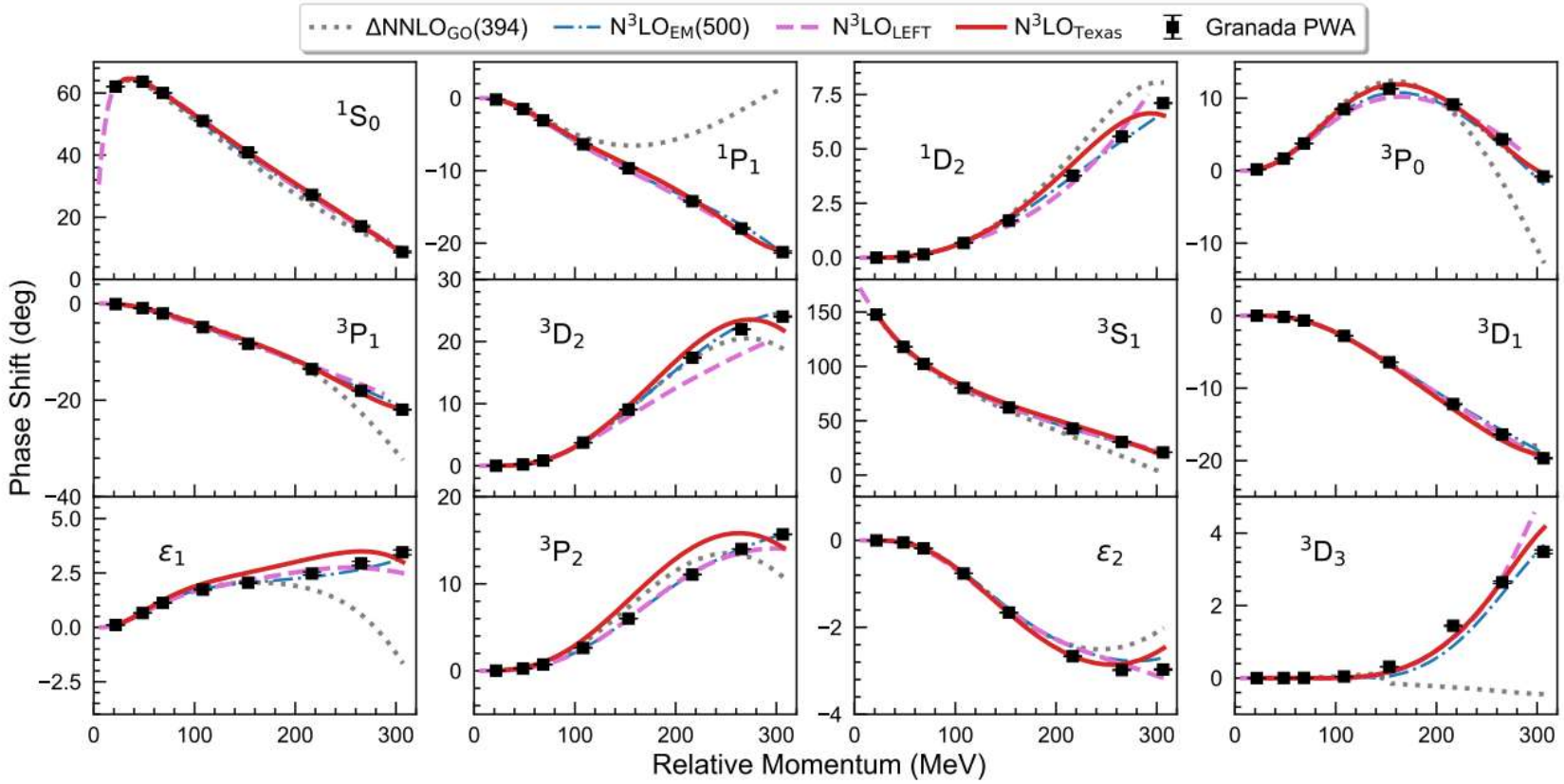
Saturation is the minimum

Symmetric matter has a minimum near $\rho_0 \approx 0.158$ fm^{-3} and $E/A \approx -16.5$ MeV (visual estimates).

Neutron-matter energy rises with density over the displayed range, without the same self-bound minimum.

Nuclear matter was not fitted. The pink ellipse is an external 95% inference region.

Figure 5: scattering phase shifts and trade-offs



$$^{2S+1}L_J, \qquad k\cot\delta_0 = -1/a + (r_e/2)k^2 + \mathcal{O}(k^4)$$

S, L, J: spin, orbital and total angular momentum. a/re: scattering length/range. $\varepsilon_1/\varepsilon_2$: mixing angles.

$^3P_2$ phase shifts compete with the ^{16}O radius in fitting.

Table I: calibration observables

		N^3LO_{Texas}	Exp.
Optimization	a_{pp}^C	−7.820	−7.8196(26)[a]
	r_{pp}^C	2.757	2.790(14)[a]
	a_{nn}	−18.950	−18.95(40)
	r_{nn}	2.794	2.75(11)
	a_{np}	−23.741	−23.740(20)
	r_{np}	2.681	2.77(5)
	$E_{gs}(^2H)$	−2.225	−2.224575(9)
	$R_{ch}(^2H)$	2.131	2.1421(88)
	$Q(^2H)$	0.27	0.27[b]
	$P_D(^2H)$	2.68%	–
	$E_{gs}(^4He)$	−28.61	−28.2957
	$R_{ch}(^4He)$	1.675	1.6775(28)
	$E_{gs}(^{16}O)$	−126.54	−127.6193
	$R_{ch}(^{16}O)$	2.707	2.6991(52)

TABLE I. $^1S_0$ effective range parameters (a, r) in fm, ground-state energies (E_{gs}) in MeV, charge radii (R_{ch}) in fm, and quadrupole moment (Q) in fm^2 used to optimized the LECs. Results based on J-NCSM for $A = 2, 3, 4$, EOM-CCSDT-3 for ^{22,24}O, and IMSRG(3f2) for ^{78}Ni, ^{132}Sn and ^{208}Pb. Results from the validation step are computed with CCSDT-3. The uncertainties in the EOM-CCSDT-3 computations stem from the truncated model space. Experiment from [12, 13, 42, 78, 98–101].

Calibration data

Scattering parameters, ²H, ⁴He and ¹⁶O constrain the interaction.

The table also reports PD. It is not among the listed fitting targets.

[a] Including Coulomb (C) effects. Tuned separately after optimization.

[b] CD-Bonn value [102]

[c] Trends from mass surface [13]

Source: original Table I, upper section, End Matter p. 9. Caption and footnotes reproduced from the paper.

Table I: selection and prediction

		N^3LO_{Texas}	Exp.
Validation	$E_{gs}(^{22}O)$	−162.06	−162.0272(572)
	$E_{gs}(^{24}O)$	−168.77	−168.9525(1680)
	$E_{gs}(^{40}Ca)$	−339.15	−342.0522
	$E_{gs}(^{48}Ca)$	−417.63	−416.0012
Prediction	$E_{gs}(^{3}H)$	−8.50	−8.4818
	$R_{ch}(^{3}H)$	1.769	1.7591(363)
	$E_{gs}(^{3}He)$	−7.76	−7.7180
	$R_{ch}(^{3}He)$	1.967	1.9661(30)
	$E_{2+}(^{22}O)$	3.28(7)	3.199
	$E_{2+}(^{24}O)$	4.05(11)	4.760
	$E_{gs}(^{78}Ni)$	−647(2)	−642.5640(3900)[c]
	$R_{ch}(^{78}Ni)$	3.961	–
	$E_{gs}(^{132}Sn)$	−1102(5)	−1102.8432(19)
	$R_{ch}(^{132}Sn)$	4.723	4.7093(76)
	$E_{gs}(^{208}Pb)$	−1613(21)	−1636.4302(12)
	$R_{ch}(^{208}Pb)$	5.530	5.5012(13)

[a] Including Coulomb (C) effects. Tuned separately after optimization.

[b] CD-Bonn value [102]

[c] Trends from mass surface [13]

Candidate selection

The four energies under “Validation” select the final interaction from the 20 candidates.

Further predictions

The remaining rows test other nuclei or observables. The ^{24}O 2^+ energy remains about 15% below the listed value.

Source: original Table I, lower section, End Matter p. 9. Original header repeated for the continuation.

Table S1: fixed constants and lower-order contacts

TABLE S1. Low-energy constants (LECs) for the N^3LO_{Texas} interaction. The constants c_i, $\bar{d}_i$, $\widetilde{C}_i$, C_i, and D_i (also $\widehat{D}_i$) are in units of GeV^{-1}, GeV^{-2}, 10^4 GeV^{-2}, 10^4 GeV^{-4}, and 10^4 GeV^{-6}, respectively. The 3N LECs c_D and c_E are dimensionless.

LEC	Value	LEC	Value
	πN		
c_1	−1.07	c_2	3.20
c_3	−5.32	c_4	3.56
$\bar{d}_1 + \bar{d}_2$	1.04	$\bar{d}_3$	−0.48
$\bar{d}_5$	0.14	$\bar{d}_{14} - \bar{d}_{15}$	−1.90
	2N (LO)		
$\widetilde{C}^{pp}_{^1S_0}$	−0.154291	$\widetilde{C}^{np}_{^1S_0}$	−0.155631
$\widetilde{C}^{nn}_{^1S_0}$	−0.155010	$\widetilde{C}_{^3S_1}$	−0.202464
	2N (NLO)		
$C_{^1S_0}$	2.514273	$C_{^3P_0}$	1.025413
$C_{^1P_1}$	0.082796	$C_{^3P_1}$	−0.977112
$C_{^3S_1}$	1.075598	$C_{^3S_1-{}^3D_1}$	0.528490
$C_{^3P_2}$	−0.960907		

Fixed πN inputs

The long-range constants use Roy–Steiner central values.

Optimized 2N contacts

LO and NLO entries are shown here. Higher-order contacts continue on the next slide.

Source: original Table S1, upper section, Supplement p. 3 (PDF p. 13). The original caption specifies the units.

Table S1: N³LO and three-nucleon contacts

TABLE S1. Low-energy constants (LECs) for the N^3LO_{Texas} interaction. The constants c_i, $\bar{d}_i$, $\widetilde{C}_i$, C_i, and D_i (also $\widehat{D}_i$) are in units of GeV^{-1}, GeV^{-2}, 10^4 GeV^{-2}, 10^4 GeV^{-4}, and 10^4 GeV^{-6}, respectively. The 3N LECs c_D and c_E are dimensionless.

LEC	Value	LEC	Value
	2N (N^3LO)		
$\widehat{D}_{^1S_0}$	−1.643852	$D_{^1S_0}$	−19.286790
$D_{^3P_0}$	5.607692	$D_{^1P_1}$	10.529344
$D_{^3P_1}$	5.427348	$\widehat{D}_{^3S_1}$	5.715113
$D_{^3S_1}$	−40.129215	$D_{^3D_1}$	−4.622881
$\widehat{D}_{^3S_1-{}^3D_1}$	2.852185	$D_{^3S_1-{}^3D_1}$	0.505938
$D_{^1D_2}$	−2.369999	$D_{^3D_2}$	−4.984577
$D_{^3P_2}$	6.7635540	$D_{^3P_2-{}^3F_2}$	0.188085
$D_{^3D_3}$	−1.196276		
	3N (N^2LO)		
c_D	−5.060563	c_E	−1.034629

15 N³LO contacts

These complete the 26 two-nucleon contact parameters.

cD and cE

Two additional 3N parameters give 28 in total.

Source: original Table S1, lower section, Supplement p. 3 (PDF p. 13). Original column header repeated.

Table S2: initial parameter search domains

TABLE S2. Initial search domains for the LEC values, where $\widetilde{C}_i$, C_i, and D_i (also $\widehat{D}_i$) are in units of 10^4 GeV^{-2}, 10^4 GeV^{-4}, and 10^4 GeV^{-6}, respectively. The 3N LECs c_D and c_E are dimensionless.

LEC	Min	Max	LEC	Min	Max
$\widetilde{C}^{pp}_{^1S_0}$	−0.30	−0.10	$\widetilde{C}^{np}_{^1S_0}$	−0.30	−0.10
$\widetilde{C}^{nn}_{^1S_0}$	−0.30	−0.10	$\widetilde{C}_{^3S_1}$	−0.30	−0.10
$C_{^1S_0}$	2.30	2.90	$C_{^3P_0}$	0.80	1.40
$C_{^1P_1}$	−1.00	1.00	$C_{^3P_1}$	−1.40	1.50
$C_{^3S_1}$	0.10	1.70	$C_{^3S_1-{}^3D_1}$	0.00	1.20
$C_{^3P_2}$	−2.10	2.40	$\widehat{D}_{^1S_0}$	−2.20	4.50
$D_{^1S_0}$	−28.00	−18.00	$D_{^3P_0}$	3.90	6.40
$D_{^1P_1}$	8.50	18.00	$D_{^3P_1}$	2.00	10.50
$\widehat{D}_{^3S_1}$	−9.70	9.30	$D_{^3S_1}$	−40.60	16.70
$D_{^3D_1}$	−8.20	4.50	$\widehat{D}_{^3S_1-{}^3D_1}$	−10.00	9.30
$D_{^3S_1-{}^3D_1}$	−7.40	10.00	$D_{^1D_2}$	−2.60	1.10
$D_{^3D_2}$	−8.00	−2.00	$D_{^3P_2}$	−3.00	15.30
$D_{^3P_2-{}^3F_2}$	−2.60	2.90	$D_{^3D_3}$	−3.00	1.00
c_D	−8.00	8.00	c_E	−8.00	8.00

Initial search region

The 2000 Latin-hypercube starting points sample the parameter intervals in this table.

Final values and bounds

Table S1 gives the final values. Table S2 gives the initial minimum and maximum for each parameter.

Uncertainty and the limits of the evidence

Red bands in Figs. 1/2	Mainly selected many-body errors. Full interaction, parameter, and near-threshold effects are not covered.
<1% in Fig. S2	Emulator discrepancy relative to full calculations, not total accuracy against experiment.
CC–VS-IMSRG agreement	Supports the trend, while both solvers share uncertainty from the same interaction.
The ^{71}Ca dripline label	A theoretical S_2n threshold. One-neutron stability must be assessed separately.

A full Bayesian uncertainty budget, higher-order 3N forces, and threshold effects remain future work.

Conclusions and discussion

The new interaction yields binding beyond N = 40

Joint constraints on few-body data, energies, and radii change calcium predictions far from stability.

Two-neutron threshold near ^{71}Ca; even-A limit near ^{70}Ca

CC and VS-IMSRG support the extension. This does not imply stability of ^{71}Ca against one-neutron emission.

Tests across systems extend the evidence

Oxygen, spectra, radii, and matter provide complementary tests, with residual phase-shift and spectral discrepancies.